Martinez, Thomas — Research Proposal

1. Introduction

My research combines **combinatorics and geometry** to study flag varieties, cluster algebras, and link invariants. I use diagrams to calculate geometric invariants and geometry to explain combinatorial patterns. At UC Davis, I propose to work with Roger Casals to extend connections among these subjects from finite to affine type A. The fellowship would combine my affine Deodhar diagrams with his geometric methods to pursue two objectives: constructing **cluster coordinates from diagrams** and recovering **geometric invariants through link topology**.

2. Background and preparation

Flags are nested sequences of vector subspaces; Richardson varieties describe flags with prescribed relative positions. Their affine counterparts use periodic chains of lattices in a Laurent-series vector space. Concrete examples come from *open positroid varieties*, which parametrize subspaces with prescribed cyclic rank conditions [Pos06, KLS13]. Snider's stratification-preserving isomorphism identifies their nonempty Plücker patches, where a chosen maximal minor is nonzero, with affine open Richardson varieties [Sni11, Theorem 15]:

$$\Pi_f^\circ \cap \{\Delta_I \neq 0\} \cong \mathring{R}_{f,t_I}.$$

Here f indexes the positroid and t_I specifies the patch. An arbitrary Plücker coordinate need not occur in a seed represented by a planar graph [GL23b, OPS15], but periodic diagrams offer a way to study patches beyond these charts.

My **affine Deodhar diagrams**, periodic strand diagrams of crossings and elbows, encode the Deodhar strata of positroid patches: pieces isomorphic to $(\mathbb{C}^*)^{e(D)} \times \mathbb{C}^{b(D)}$ [1]. This turns geometric decompositions into combinatorial data. With Matthew J. Tyler, I prove that an affine type-A open Richardson variety is an algebraic torus exactly when its Deodhar decomposition has one stratum. In finite type A, we also prove that a nonempty braid Richardson variety has a dense orbit under its standard torus action exactly when its associated link is *smoothly slice* [2, Theorems 1.1 and 1.4]: its components bound disjoint smooth disks in the four-ball.

My manuscript [3], in preparation, interprets rational parking-function numbers as sums of Euler characteristics of torus quotients of open positroid varieties. It also gives direct diagrammatic bijections with parking functions, labeled lattice paths, in the Fuss–Catalan case. These results extend the geometric perspective on Catalan enumeration developed by Galashin–Lam [GL24], motivating the search for geometric explanations of refined combinatorial counts.

3. Research plan

3.1. **Geometric cluster structures.** A *cluster algebra* is generated by variables grouped into seeds and related by *mutation*, with exchange rules encoded in a directed graph called a *quiver*. Frozen variables are never mutated and are inverted here. The *upper cluster algebra* consists of functions Laurent in every seed. Bao–Ye identify affine open Richardson coordinate rings with upper cluster algebras and conjecture that the associated cluster algebras are locally acyclic [BY25, Theorem 5.13 and Conjecture 5.15]. This covering property implies equality of the two algebras [Mul13].

Numbered references indicate my work, including joint work.

In work in progress with Moriah Elkin and Tyler, we construct candidate quivers from affine 3D plabic graphs and prove that their cluster algebras are locally acyclic [4], extending finite constructions [GLSBS22]. Identifying these diagrammatic algebras with Bao–Ye's geometric structures would give

$$\mathbb{C}[\mathring{R}_{u,w}] = \mathcal{A}(\Sigma_{u,w}) = \mathcal{U}(\Sigma_{u,w}),$$

so cluster variables and inverses of frozen variables generate every regular function.

Scott's work on Grassmannians, followed by Galashin–Lam's work on open positroid varieties, showed that combinatorial diagrams describe entire coordinate rings [Sco06, GL23b]. Casals's joint work extends this picture to braid varieties, spaces of flag sequences prescribed by braid crossings, through geometric diagrams called *weaves*, and identifies the resulting structures with Deodhar constructions [CGGLSS25, CGGSSBS25]. An affine analogue would give periodic diagrams the same geometric role, connecting abstract coordinate-ring theorems with explicit geometry. It would extend the reach of these methods to positroid patches beyond planar seeds and make their coordinate changes accessible through diagrams.

3.2. **Links, fillings, and cohomology.** Galashin–Lam's work connects Richardson geometry with knot theory: finite-field point counts arise from the HOMFLY polynomial, while compactly supported equivariant cohomology corresponds to part of link homology [GL24, Theorems 1.17 and 2.1]. For certain open positroid varieties, their results also interpret rational q,t-Catalan polynomials through mixed Hodge theory.

I propose to associate to each affine open Richardson variety a link $L_{u,w} \subset S^3$ whose normalized highest-a-degree HOMFLY coefficient satisfies

$$P^{\mathrm{top}}_{L_{u,w}}(q) \stackrel{?}{=} \#\mathring{R}_{u,w}(\mathbb{F}_q) = \sum_D (q-1)^{e(D)} q^{b(D)}.$$

The second equality is the established Deodhar point-count formula [BD94, 2]. The proposed first equality would extend the finite-type correspondence to affine Richardson varieties. For positroid patches, where $(u,w) = (f, t_I)$, it would give a topological interpretation of point counts that depend on the chosen Plücker coordinate.

A natural starting point is the affine braid $\beta(w)\beta(u)^{-1}$, where β denotes the reduced positive braid lift. This braid lives on an annulus; closing its braid direction produces a link in $T^2 \times [0,1]$ [GN23, §1.1]. Embedding this thickened torus into S^3 gives candidate links for the proposed correspondence. Affine Hecke traces and monotone links provide precedents for relating this periodic geometry to link invariants [GN23, GL23a].

A complementary approach comes from contact and symplectic geometry. Up to algebraic torus factors, open positroid varieties admit models as augmentation varieties, parameter spaces associated to Legendrian links [CGGS26, Theorems 1.2–1.3]. For positive-braid rainbow closures, every seed of the augmentation variety with one marked point per component is realized by an embedded exact Lagrangian filling [CG24, Corollary 1.3], a bounding surface that gives geometric meaning to the cluster chart. An affine analogue could interpret mutations through changes of bounding surfaces, connecting my diagrams with Casals's symplectic methods. A homological refinement of such invariants could recover mixed Hodge gradings and give geometric meaning to refined counts suggested by my parking-function work.

4. UC Davis and fellowship development

UC Davis is an especially well-equipped university to attain these research goals. In addition to discussions with and the mentorship of Roger Casals, discussions with Melissa Sherman-Bennett on braid variety cluster structures would inform the coordinate constructions [GLSBS22, CGGSSBS25]. Eugene Gorsky's work on affine Hecke traces [GN23] and Melissa Zhang's work on annular Khovanov homology [Zha18] would offer complementary perspectives on the proposed link invariants.

I would participate in the Algebra & Discrete Mathematics and Geometry/Topology seminars and help organize working groups on these affine questions. At UCLA, I have helped organize the Combinatorics Research Seminar and co-organized a learning seminar on Coxeter groups and root systems. I would also build on my undergraduate mentoring at UCLA by proposing student research projects through the Davis Math Lab.

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